

Description about conservative horizontal momentum diffusion scheme and tensorial stress option in LFRic

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1 Equation of diffusion scheme

In LFRic's Smagorinsky scheme and blending boundary layer scheme [1], subgrid turbulent mixing of momentum is calculated by the following equation:

$$\frac{\partial u_i}{\partial t} = -\frac{1}{\rho} \frac{\partial F^{u_i, x_j}}{\partial x_j} \quad \left(F^{u_i, x_j} = -\rho \nu \frac{\partial u_i}{\partial x_j} \right), \quad (1)$$

where ρ and ν are density and mixing coefficient, respectively. u_i and x_i are 3-dimensional cartesian velocity and coordinate following Einstein's notation rule. Incompressible formulation is adopted, so density is ignored in the horizontal diffusion, so each F^{u_i, x_j} is represented as follows:

$$\begin{aligned} F^{ux} &= -\nu \frac{\partial u}{\partial x}, & F^{uy} &= -\nu \frac{\partial u}{\partial y}, & F^{uz} &= -\nu \frac{\partial u}{\partial z}, \\ F^{vx} &= -\nu \frac{\partial v}{\partial x}, & F^{vy} &= -\nu \frac{\partial v}{\partial y}, & F^{vz} &= -\nu \frac{\partial v}{\partial z}, \\ F^{wx} &= -\nu \frac{\partial w}{\partial x}, & F^{wy} &= -\nu \frac{\partial w}{\partial y}, & F^{wz} &= -\nu \frac{\partial w}{\partial z}, \end{aligned}$$

where x, y, z, u, v and w are 3-dimensional coordinate and wind components, respectively. Since, red colored terms are calculated not in the diffusion scheme but in the boundary layer scheme, we ignore them in the following explanation. The equations for u, v, w are as follows:

$$\frac{\partial u}{\partial t} = -\frac{\partial F^{ux}}{\partial x} - \frac{\partial F^{uy}}{\partial y}, \quad (2)$$

$$\frac{\partial v}{\partial t} = -\frac{\partial F^{vx}}{\partial x} - \frac{\partial F^{vy}}{\partial y}, \quad (3)$$

$$\frac{\partial w}{\partial t} = -\frac{\partial F^{wx}}{\partial x} - \frac{\partial F^{wy}}{\partial y}. \quad (4)$$

2 Conservative horizontal diffusion

Here, we will explain how to calculate (2), (3) and (4). In the following, as shown in Figure1, locations that are half-grid shifted relative to grid numbers i, j and k will be represented as $i + \frac{1}{2}, j + \frac{1}{2}$ and $k + \frac{1}{2}$, respectively.

In LFRic, u, v and w are placed on the face of each cell as shown in Figure2.

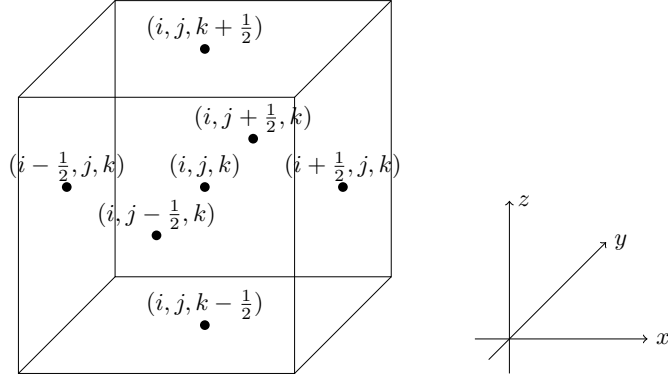


Figure1: Notation of grid number. Cube indicates (i, j, k) th cell.

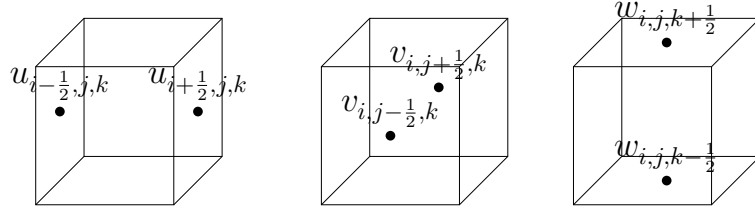


Figure2: Definition point of each velocity on the face of (i, j, k) th cell.

In the calculation of (2) for $u_{i+\frac{1}{2}, j, k}$, we assume a volume $V_{i+\frac{1}{2}, j, k}$ surrounding $u_{i+\frac{1}{2}, j, k}$. ν is linearly interpolated to the face of $V_{i+\frac{1}{2}, j, k}$, and wind shears are also evaluated at the same locations. F^{ux} and F^{uy} are calculated on the face of $V_{i+\frac{1}{2}, j, k}$ by multiplying them as depicted in Figure3. Increment of $u_{i+\frac{1}{2}, j, k}$ is calculated as:

$$\Delta u_{i+\frac{1}{2}, j, k} = - \frac{\left(F^{ux}_{i+1, j, k} A^x_{i+1, j, k} - F^{ux}_{i, j, k} A^x_{i, j, k} \right) + \left(F^{uy}_{i, j+\frac{1}{2}, k} A^y_{i, j+\frac{1}{2}, k} - F^{uy}_{i, j-\frac{1}{2}, k} A^y_{i, j-\frac{1}{2}, k} \right)}{V_{i+\frac{1}{2}, j, k}} \Delta t, \quad (5)$$

where A^x and A^y are the area of faces on which F^{ux} and F^{uy} are placed, respectively. Please see Figure3 for placement of each term in (5). Since adjacent cells share the same $F^{ux}A^x$ and $F^{uy}A^y$ at their boundary, the increment of total volume-weighted u is cancelled out by summing up (5) and ignoring the effect of lateral boundary condition:

$$\Delta \left(\sum_{i, j, k} V_{i+\frac{1}{2}, j, k} u_{i+\frac{1}{2}, j, k} \right) = \sum_{i, j, k} V_{i+\frac{1}{2}, j, k} \Delta u_{i+\frac{1}{2}, j, k} = 0, \quad (6)$$

which means (5) conserves volume-weighted total u .

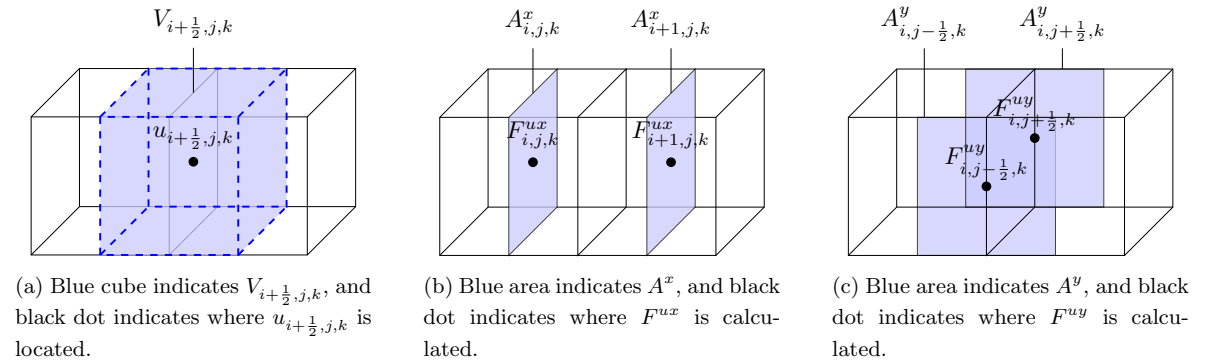


Figure3: Placement of each terms appered in (5).

In the calculation of $\Delta v_{i,j+\frac{1}{2},k}$ and $\Delta w_{i,j,k+\frac{1}{2}}$, we assume volumes $V_{i,j+\frac{1}{2},k}$ and $V_{i,j,k+\frac{1}{2}}$, respectively, and calculate F^{vx} , F^{vy} , F^{wx} , F^{wy} , A^x and A^y on their faces. Increments of $v_{i,j+\frac{1}{2},k}$ and $w_{i,j,k+\frac{1}{2}}$ are calculated in the same manner as (5):

$$\Delta v_{i,j+\frac{1}{2},k} = - \frac{\left(F_{i+\frac{1}{2},j,k}^{vx} A_{i+\frac{1}{2},j,k}^x - F_{i-\frac{1}{2},j,k}^{vx} A_{i-\frac{1}{2},j,k}^x\right) + \left(F_{i,j+1,k}^{vy} A_{i,j+1,k}^y - F_{i,j,k}^{vy} A_{i,j,k}^y\right)}{V_{i,j+\frac{1}{2},k}} \Delta t, \quad (7)$$

$$\Delta w_{i,j,k+\frac{1}{2}} = - \frac{\left(F_{i+\frac{1}{2},j,k+\frac{1}{2}}^{wx} A_{i+\frac{1}{2},j,k+\frac{1}{2}}^x - F_{i-\frac{1}{2},j,k+\frac{1}{2}}^{wx} A_{i-\frac{1}{2},j,k+\frac{1}{2}}^x\right) + \left(F_{i,j+\frac{1}{2},k+\frac{1}{2}}^{wy} A_{i,j+\frac{1}{2},k+\frac{1}{2}}^y - F_{i,j-\frac{1}{2},k+\frac{1}{2}}^{wy} A_{i,j-\frac{1}{2},k+\frac{1}{2}}^y\right)}{V_{i,j,k+\frac{1}{2}}} \Delta t, \quad (8)$$

Please see Figure4 and Figure5 for placement of each term in (7) and (8), respectively. Similarly, volume-weighted total v and w are conserved through the calculation.

$$\Delta \left(\sum_{i,j,k} V_{i,j+\frac{1}{2},k} v_{i,j+\frac{1}{2},k} \right) = \sum_{i,j,k} V_{i,j+\frac{1}{2},k} \Delta v_{i,j+\frac{1}{2},k} = 0 \quad (9)$$

$$\Delta \left(\sum_{i,j,k} V_{i,j,k+\frac{1}{2}} w_{i,j,k+\frac{1}{2}} \right) = \sum_{i,j,k} V_{i,j,k+\frac{1}{2}} \Delta w_{i,j,k+\frac{1}{2}} = 0 \quad (10)$$

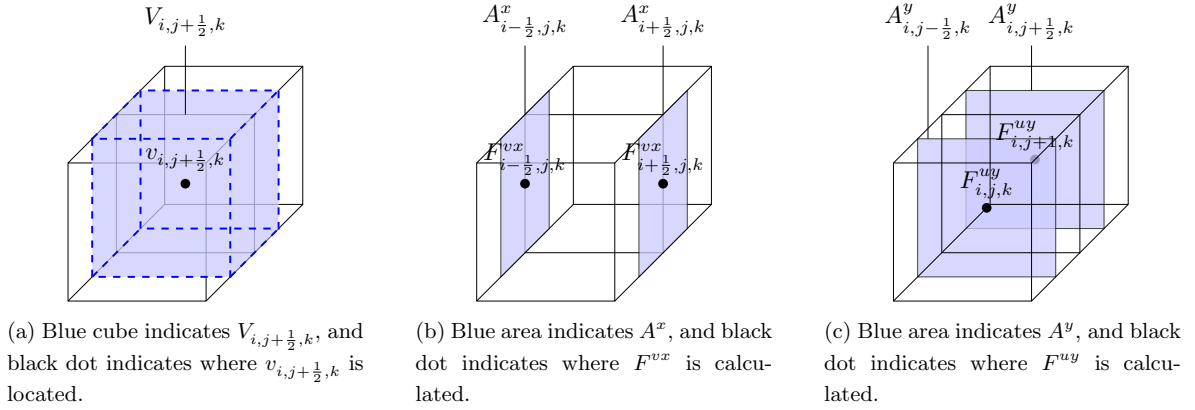


Figure4: Placement of each terms appeared in (7).

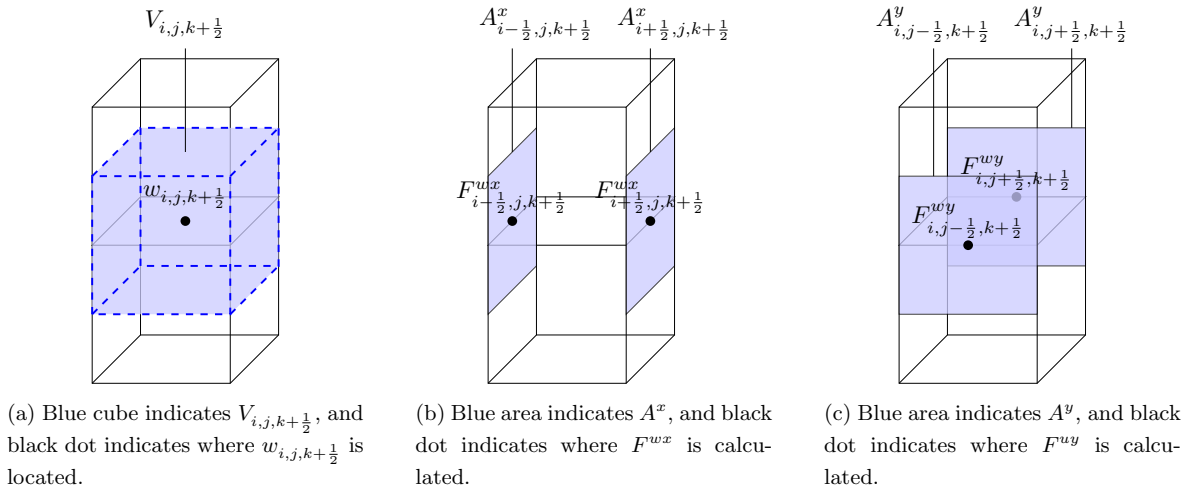


Figure5: Placement of each terms appeared in (8).

3 Tensorial momentum diffusion option

Here, we will explain the equation of tensorial diffusion equation and how to calculate it. When tensorial stress is used, (1) will be:

$$\frac{\partial u_i}{\partial t} = -\frac{1}{\rho} \frac{\partial F^{u_i, x_j}}{\partial x_j} \quad \left(F^{u_i, x_j} = -\rho\nu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right), \quad (11)$$

where blue colored terms are newly added in the tensorial diffusion. Adopting incompressible formulation, each $F^{u_i x_j}$ is represented as:

$$\begin{aligned} F^{ux} &= -2\nu \frac{\partial u}{\partial x}, & F^{uy} &= -\nu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), & F^{uz} &= -\nu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right), \\ F^{vx} &= -\nu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), & F^{vy} &= -2\nu \frac{\partial v}{\partial y}, & F^{vz} &= -\nu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right), \\ F^{wx} &= -\nu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right), & F^{wy} &= -\nu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right), & F^{wz} &= -2\nu \frac{\partial w}{\partial z}. \end{aligned}$$

Ignoring red colored terms, which are calculated not in the diffusion scheme but in the boundary layer scheme, the equations for u, v and w will be:

$$\frac{\partial u}{\partial t} = -\frac{\partial F^{ux}}{\partial x} - \frac{\partial F^{uy}}{\partial y} - \frac{\partial F^{uz*}}{\partial z} \quad \left(F^{uz*} = -\nu \frac{\partial w}{\partial x} \right), \quad (12)$$

$$\frac{\partial v}{\partial t} = -\frac{\partial F^{vx}}{\partial x} - \frac{\partial F^{vy}}{\partial y} - \frac{\partial F^{vz*}}{\partial z} \quad \left(F^{vz*} = -\nu \frac{\partial w}{\partial y} \right), \quad (13)$$

$$\frac{\partial w}{\partial t} = -\frac{\partial F^{wx}}{\partial x} - \frac{\partial F^{wy}}{\partial y}. \quad (14)$$

We can calculate the first and the second terms on the right hand side of (12), (13) and (14) in the same way as (5), (7) and (8), respectively, except for adding blue colored terms in $F^{u_i x_j}$. In the discretization of the third terms in (12), (13), we adopt a similar way so that it still conserves volume-weighted total u and v :

$$-\frac{\partial F^{uz*}}{\partial z} \Big|_{i+\frac{1}{2}, j, k} = -\frac{F^{uz*}_{i+\frac{1}{2}, j, k+\frac{1}{2}} A^z_{i+\frac{1}{2}, j, k+\frac{1}{2}} - F^{uz*}_{i+\frac{1}{2}, j, k-\frac{1}{2}} A^z_{i+\frac{1}{2}, j, k-\frac{1}{2}}}{V_{i+\frac{1}{2}, j, k}} \quad (15)$$

$$-\frac{\partial F^{vz*}}{\partial z} \Big|_{i, j+\frac{1}{2}, k} = -\frac{F^{vz*}_{i, j+\frac{1}{2}, k+\frac{1}{2}} A^z_{i, j+\frac{1}{2}, k+\frac{1}{2}} - F^{vz*}_{i, j+\frac{1}{2}, k-\frac{1}{2}} A^z_{i, j+\frac{1}{2}, k-\frac{1}{2}}}{V_{i, j+\frac{1}{2}, k}} \quad (16)$$

Please see Figure6 for placement of each term in (15) and (16).

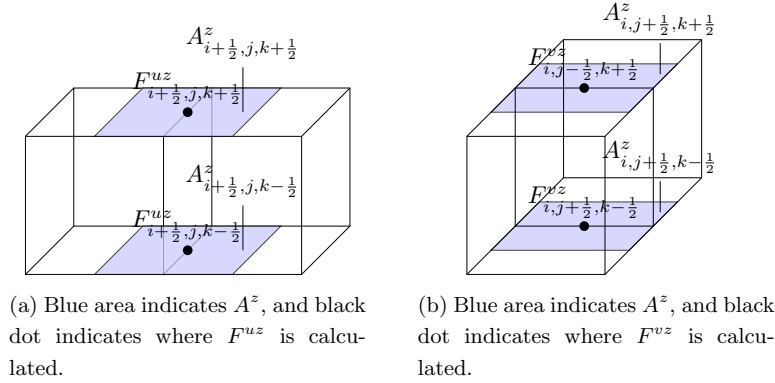


Figure6: Placement of each terms appered in (15) and (16).

4 Usage of function space

In the LFRic's source code, we use $\mathbb{W}_3$ function spaces for variables placed at the center of the cell, while we use $\mathbb{W}_1$ or shifted $\mathbb{W}_2^H$ function space for variables placed on the edges of the cell. Each degree of freedom (dof) in $\mathbb{W}_1$ and shifted $\mathbb{W}_2^H$ function space is placed as shown in Figure7. Function spaces used for each $F^{u_{ij}}$ is listed in table1. As shown in table1, we sometimes treat 2 variables together in a single variable (e.g. we treat F^{uy} and F^{uz*} in a single variable `uflux.w1`).

Notation in this document	Placement	Variable name in the source code	function space
F^{ux}	(i, j, k)	<code>uflux.w3</code>	$\mathbb{W}_3$
F^{uy}	$(i + \frac{1}{2}, j + \frac{1}{2}, k)$	<code>uflux.w1</code>	$\mathbb{W}_1$
F^{uz*}	$(i + \frac{1}{2}, j, k + \frac{1}{2})$		
F^{vy}	(i, j, k)	<code>vflux.w3</code>	$\mathbb{W}_3$
F^{vx}	$(i + \frac{1}{2}, j + \frac{1}{2}, k)$	<code>vflux.w1</code>	$\mathbb{W}_1$
F^{vz*}	$(i, j + \frac{1}{2}, k + \frac{1}{2})$		
F^{wx}	$(i + \frac{1}{2}, j, k + \frac{1}{2})$	<code>wflux.sh.w2h</code>	shifted $\mathbb{W}_2^H$
F^{wy}	$(i, j + \frac{1}{2}, k + \frac{1}{2})$		

Table1: Placement and function space for each variable.

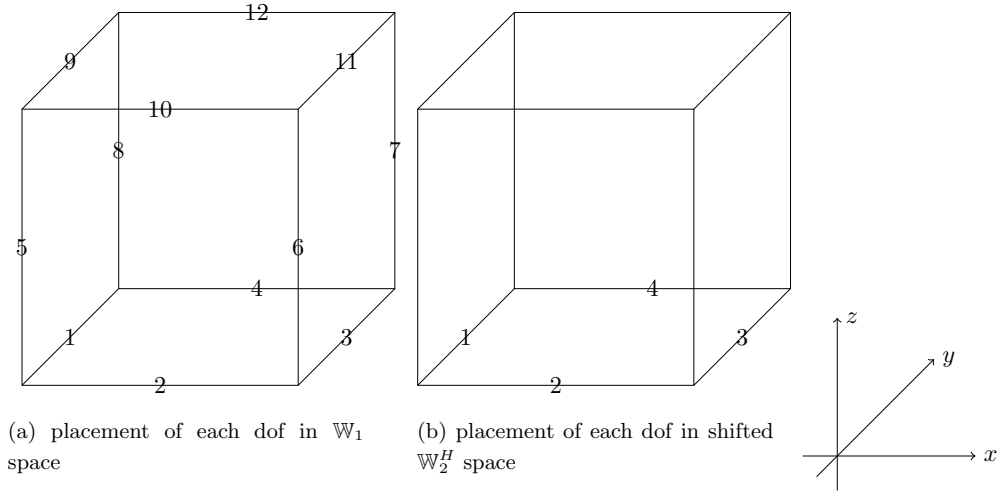


Figure7: placement of dof

5 Point to note

5.1 Rigor in other coordinates

Equations solved in this scheme holds in only cartesian coordinate, but LFRic also uses other coordinates such as cubes-sphere and rotated latitude-longitude coordinates. Strictly speaking, metric terms such as metric tensor and Christoffel symbols must be taken into account in other coordinates.

5.2 Non-conservation in nesting run.

This scheme is designed to calculate increments of outer most wind as 0 in the nesting run. This design is natural since outer most wind should be always equal as that of parent model in the nesting run. But we have to note this rigid boundary condition breaks conservation of volume-weighted total u, v and w in nesting run while they are conserved rigorously in idealised simulation using periodic lateral boundary condition. For example, given $u_{i+\frac{1}{2},j,k}$ is located at the lateral boundary in y -direction as depicted in Figure8, we implicitly assume $F_{i+\frac{1}{2},j+\frac{1}{2}}^{uy}$, the lateral boundary condition of u , so that it satisfies:

$$\Delta u_{i+\frac{1}{2},j,k} = -\frac{\partial F^{ux}}{\partial x}_{i+\frac{1}{2},j,k} - \frac{\partial F^{uy}}{\partial y}_{i+\frac{1}{2},j,k} - \frac{\partial F^{uz*}}{\partial z}_{i+\frac{1}{2},j,k} = 0, \quad (17)$$

that is:

$$\begin{aligned} F_{i+\frac{1}{2},j+\frac{1}{2},k}^{uy} A_{i+\frac{1}{2},j+\frac{1}{2},k}^y &= F_{i+\frac{1}{2},j-\frac{1}{2},k}^{uy} A_{i+\frac{1}{2},j-\frac{1}{2},k}^y - (F_{i+1,j,k}^{ux} A_{i+1,j,k}^x - F_{i,j,k}^{ux} A_{i,j,k}^x) \\ &\quad - \left(F_{i+\frac{1}{2},j,k+\frac{1}{2}}^{uz*} A_{i+\frac{1}{2},j,k+\frac{1}{2}}^z - F_{i+\frac{1}{2},j,k-\frac{1}{2}}^{uz*} A_{i+\frac{1}{2},j,k-\frac{1}{2}}^z \right) \end{aligned} \quad (18)$$

This non-zero term resides after summing up (5) and changes the total volume-weighted u .

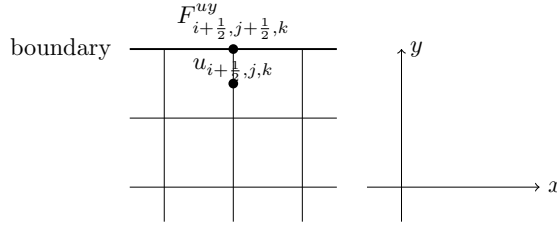


Figure8: Horizontal cross section close to the lateral boundary

References

- [1] I. A. Boutle, J. E. Eyre, and A. Lock. Seamless stratocumulus simulation across the turbulent gray zone. *Monthly Weather Review*, 142(4):1655–1668, 2014.